

TEAM SAS CODIES

SBI Saarthi

Your bank, one step ahead.

A bank waits for you to open the app, blasts the same offer to everyone, then leaves you to do the work. For SBI's **50 crore customers**, a relationship manager exists only for the wealthiest 1%. **Saarthi turns engagement into a crew of autonomous AI agents** that sense each customer's money-moment, reach them in their **own language**, and **complete the action for them** — book the FD, shift the EMI, revive the account — every step consent-gated, guardrail-checked and audit-logged.

50 crore

CUSTOMERS SBI SERVES

~1%

GET A HUMAN RELATIONSHIP
MANAGER

22

LANGUAGES SAARTHI SPEAKS ·
VOICE & TEXT

Engagement that waits, guesses, and offloads the work.

Digital engagement today is **reactive**, **generic** and **unactionable**. At SBI's scale, that's hundreds of crores of relationships reduced to a notification nobody opens.

Reactive

It waits for the customer to open YONO. Most never go beyond a balance check — the bank never reaches out first.

Generic

Broadcast SMS & push to everyone — notification blindness, collapsing CTR, opt-outs.

Unactionable

It gives advice, then asks the customer to do the work. They abandon before completing.

A relationship manager exists for **~1%** of customers. The other **49.5 crore** get the same broadcast SMS — engagement is an **attention-and-action** problem, not a data one.

01

Signal

Salary lands; ₹85k sits idle for weeks.



02

Broadcast

A generic "open an FD" SMS goes to millions.



03

Ignored

Wrong time, wrong language, no easy action.



04

Drift

Idle money, dormant account, churn to fintechs.

The engagement layer **nobody** has built.

FORCE 01

Agentic AI crossed the line

Reliable tool-use and multi-agent orchestration (LangGraph, Claude) now let software *act* safely on a customer's behalf — not just chat. Autonomy with guardrails is production-grade.

- From chatbots to agents that act

FORCE 02

Bharat is consent-ready

UPI-scale rails, the live Account Aggregator framework, and Bhashini's 22-language stack make vernacular, consented, real-time action possible at population scale for the first time.

- Consent · vernacular · real-time

FORCE 03

The churn clock is ticking

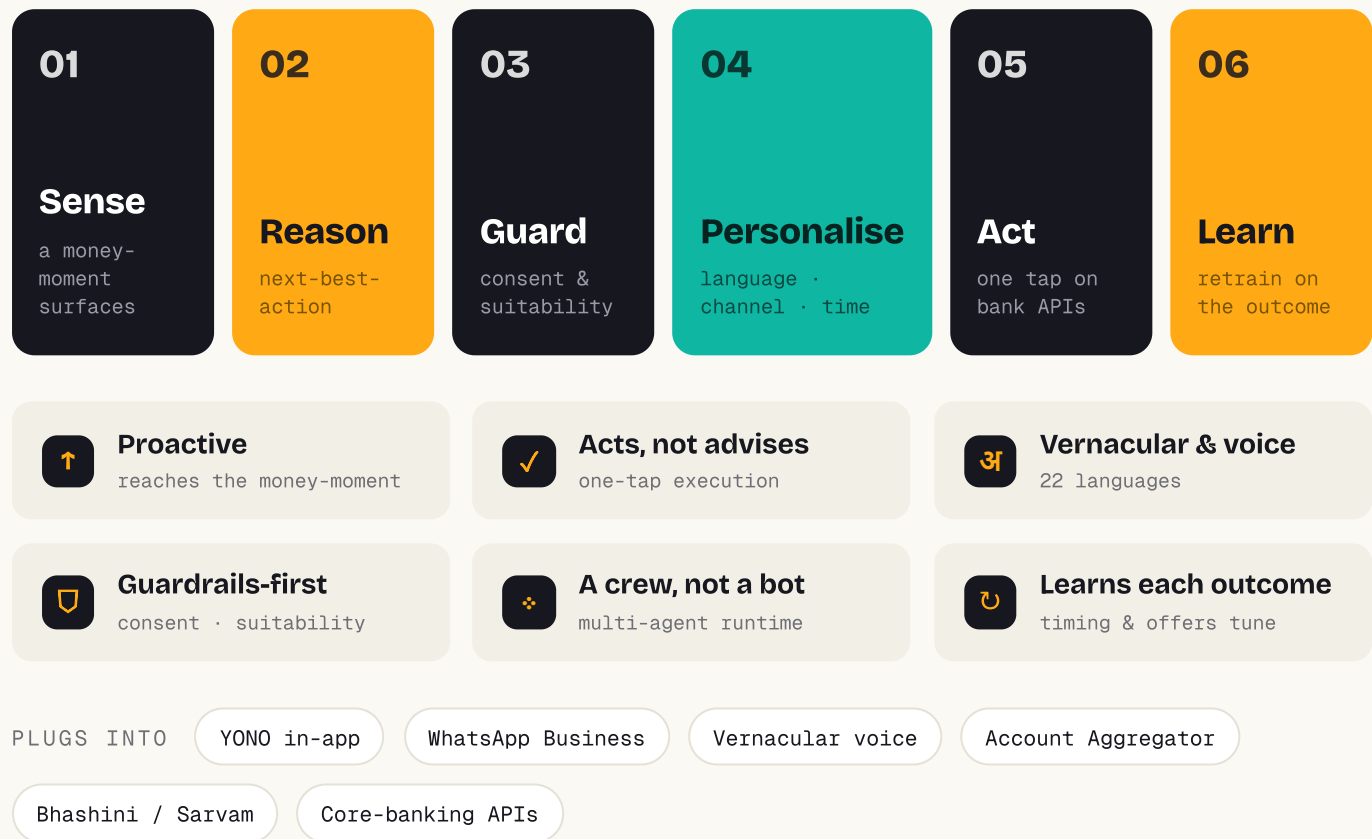
Fintechs already own daily engagement; idle CASA and dormant accounts are lifetime value bleeding out. Proactive, personal engagement is now a balance-sheet defence.

- Engagement defends the balance sheet

THE UNFILLED GAP

Banks can **detect** moments and own the **channels** — but the proactive, explainable, action-closing, vernacular engagement layer that reaches the right customer, completes the action under consent and guardrails, and learns from every outcome **does not exist**. That is Saarthi.

One loop: signal to finished action.



Beyond notifications: the closed **sense** → **reason** → **guard** → **personalise** → **act** → **learn** loop that reaches the right customer and finishes the job — for all 50 crore, not just the 1%.

From signal to a finished, audited action.

SIGNALS — REAL-TIME, CROSS-CHANNEL

Core-banking

UPI · IMPS

Balances · SIs

EMIs · bills

**Acct
Aggregator**

consented

**App
telemetry**

YONO events

Life events

salary · tax

**Product ·
policy**

catalog · RAG

**SBI SAARTHI — AGENTIC ENGAGEMENT RUNTIME**

LANGGRAPH

LAYER 01**Sensor**

detects the money-moment

- + Kafka event stream
- + idle · EMI · dormant
- + tax · scam · life
- + frequency caps

LAYER 02**Orchestrator**

plans & routes the crew

- + who · when · what
- + priority & budget
- + memory (cust-360)
- + human-in-the-loop

LAYER 03**Agent crew**

specialised, tool-using

- + Insight / NBA
- + Guardrail / consent
- + Personalization
- + Action / execution

LAYER 04**Learning**

reward from every outcome

- + contextual bandit
- + timing / channel
- + offline retrain
- + drift monitoring

**OMNICHANNEL DELIVERY**

Right channel, moment & language — YONO · WhatsApp · voice · SMS, 22 languages, frequency-capped, opt-outs honoured.

**ONE-TAP ACTION**

Executes on banking APIs — FD · SIP · EMI shift · reactivate. Consent token + maker-checker, suitability-checked, never silent.

● Consent (AA) ● PII-minimised ● Audit-logged ● Human-in-loop ● Event-driven · cloud-native

One message: sensed, reasoned, **acted.**

The customer sees only a timely message in their language with a single tap. Behind it, the crew has reasoned, cleared guardrails, and prepared a fully audited action.

S

Saarthi · Engagement Console

ILLUSTRATIVE · IDLE BALANCE

PROACTIVE · VERNACULAR · AUDITED

Ravi K. · Savings ----7731

Pune · salary credited 6 days ago ·
moment sensed 0.4s ago

0.82
PROPENSITY

WHY NOW — NEXT-BEST-ACTION REASON CODES

₹85,000 idle in savings for 40 days	+0.34
Stable salary inflow, low volatility	+0.21
Risk profile fits a liquid Sweep-FD	+0.17
Prefers Hindi · opens WhatsApp ~11:00	+0.10

MESSAGE DELIVERED — CUSTOMER'S LANGUAGE

WHATSAPP · HINDI

“नमस्ते रवि 🙌 आपके बचत खाते में ₹85,000 पिछले 40 दिनों से पड़े हैं। मैं ₹50,000 को 7.1% स्वीप-FD में लगा दूँ — इस साल लगभग ₹3,140 ज़्यादा, और पैसा कभी भी निकाला जा सकता है।”

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EN

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+19

RECOMMENDED ACTION

Open ₹50,000 Sweep-FD

7.1% · fully liquid · est. +₹3,140 this year. One tap books it on core-banking.

Yes, set it up

Not now

DECISION GOVERNANCE

Sense → NBA →

Guardrail → Consent →

Act → Audit

Suitability & consent cleared before delivery; high-value execution is maker-checker and immutably logged: who, why, when. Never spam, never silent.

Models Claude (plan/tool-use) + SLM · contextual bandit

Consent-gated · every action audit-logged

The intelligence under the **one tap**.

SIGNALS & FEEDS



Real-time · Kafka CDC
TRANSACTIONS · BALANCES · DUE DATES

Consented · AA
EXTERNAL ACCOUNTS, WITH PERMISSION

Context · RAG
PRODUCT, POLICY & CUSTOMER-360

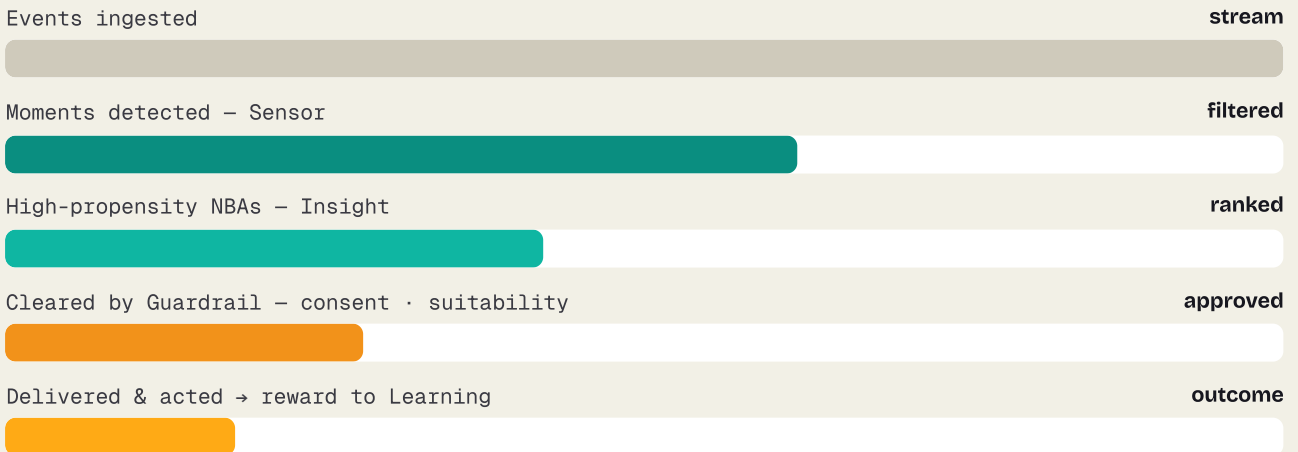
Synthetic data only for the prototype — **no real customer data**. Every feature traces to a consented, auditable source.

MODEL MIX — RIGHT MODEL FOR EACH JOB

- ▶ **Claude** — planning, tool-use, guardrails
- ▶ **Fine-tuned SLMs** — high-volume routine NBA
- ▶ **XGBoost / LightGBM** — propensity scoring
- ▶ **Contextual bandit** — timing & channel
- ▶ **Embeddings + RAG** — product / policy
- ▶ **Bhashini / Sarvam** — 22-language ASR·TTS·MT

Cost-tiered: small models handle **~90%** of routine actions; Claude is reserved for complex, high-stakes reasoning.

THE DECISION FUNNEL — RAW EVENTS TO A FINISHED ACTION



Every layer is observable and replayable — **LangSmith traces, drift monitoring and an A/B challenger** keep the crew honest and improving with every outcome.

A 30-day build to a **working prototype**.

If shortlisted, the team converts the idea into a live, demoable prototype in the ~30-day window — built on synthetic data, scoped to the hero scenarios, gated weekly.

1

WEEK 1

Foundation

LangGraph orchestrator, synthetic event stream, customer-360 personas, the Sensor agent firing on real "moments".

GATE — a moment triggers an engagement decision end-to-end.

2

WEEK 2

The crew

Insight/NBA, Guardrail/consent & Personalization agents; propensity model + contextual bandit on synthetic data.

GATE — a suitable, consent-cleared NBA, explained.

3

WEEK 3

Act & vernacular

Action agent on mock banking APIs, one-tap execution, WhatsApp + a Hindi voice demo, EN/HI/TA messaging.

GATE — a customer completes an action in their language.

4

WEEK 4

Cockpit & proof

Engagement console, impact dashboard, audit log, drift & A/B hooks; hardening & the demo script.

GATE — 5 hero scenarios run end-to-end, audited.

Synthetic data only

Consent-first

Guardrails before delivery

Vernacular by default

Everything audited

30 JUN

Round 1 — idea

15 JUL

Top 10 shortlist

JUL-AUG

30-day build

20 AUG

Jury demo

9-11 SEP

GFF · Mumbai

Engagement you can put a number on.

+34%

MONTHLY ACTIVE
ENGAGEMENT

₹1,240cr

IDLE FUNDS
MOBILISED

2.1M

DORMANT ACCOUNTS
REVIVED

+18%

CROSS-SELL
CONVERSION

01 Incremental product revenue — proactive NBAs lift cross-sell of loans, MF, insurance & deposits.

02 Idle-fund mobilisation — idle CASA converted into FDs, sweeps & SIPs.

03 Dormant revival & retention — silent accounts re-engaged; churn to fintechs defended.

04 Lower cost-to-serve — routine engagement automated vs. human RMs & call centres.

ILLUSTRATIVE IDLE-CASA LEVER – ONE MECHANISM, CONSERVATIVE ASSUMPTIONS

Of idle savings balances, a small mobilised share at a modest spread compounds across a 50-crore base:

Idle CASA pool

Mobilised

Margin uplift

Directional projections for the hackathon, not a forecast. Even a 1-2% lift across these levers is thousands of crores at SBI scale.

Saarthi runs internally as SBI's engagement layer over YONO, and productises as a licensable **Agentic Engagement Platform** for SBI subsidiaries, RRBs & co-operative banks — **per-active-customer or outcome-based pricing**.

Not a chatbot. A crew that acts.

	PROACTIVE AT THE MOMENT	PERSONALISED TO ONE	VERNACULAR + VOICE	COMPLETES THE ACTION	MULTI-AGENT + GUARDRAILS	LEARNS EACH OUTCOME
Broadcast campaigns SMS / push blasts	☐	☐	☒	☐	☐	☐
Rule-based NBA static next-best-offer	☒	☒	☐	☐	☐	☒
Support chatbot reactive Q&A	☐	☒	☒	☒	☐	☐
SBI Saarthi agentic engagement	☒	☒	☒	☒	☒	☒

Full Partial None

Each capability is something the others structurally cannot do. A campaign can't reason; a rules engine can't speak Tamil by voice and finish the action; a chatbot waits to be asked. **Saarthi is the only row that is proactive, vernacular, action-closing and guardrailed at once** — because it's a crew of autonomous agents, not a single model.

Bank-grade trust, by **design**.

■ Consent

Account Aggregator (Sahamati) — tokenised, purpose-limited, revocable.

■ Privacy

DPDP Act 2023 — PII redaction & tokenisation, data minimisation, India residency.

■ Regulatory

RBI digital-lending & outsourcing norms; SEBI suitability for any MF advice.

■ AI guardrails

OPA policy engine + LLM-as-judge; suitability, frequency caps, allow/deny actions.

■ Security

Encryption in transit & at rest, Vault/HSM secrets, RBAC, prompt-injection defence.

■ Auditability

Immutable audit trail, full agent trace, explainable decisions, human override.

Saarthi makes every one of SBI's 50 crore customers feel **personally banked — proactively, in their **own language**, with the action already done. That is Digital Engagement, finished.**

• SBI SAARTHI •

DEMO Live prototype — saudsatopay.github.io/sbi-saarthi · Code — github.com/SaudSatopay/sbi-saarthi

STACK LangGraph · Claude & SLMs · AI4Bharat/Bhashini & Sarvam · XGBoost & contextual bandits · Kafka · Postgres + pgvector · Account Aggregator · OPA

CONTEXT SBI Hackathon @ Global Fintech Fest 2026 · Agentic AI & Emerging Tech · Digital Engagement · Concept by Team SAS CODIES. Figures illustrative. Not affiliated with State Bank of India.